

¡PAPER TITLE!
Cosmological-Constant / Vacuum-Density Closure under v5.78

Star-Magic UQFF
PAPER_¡NNNN! / Session ¡SESSION!
¡DATE!

Abstract

¡2–6 sentence abstract. State: (a) the cosmological-constant/dark-energy anchor being closed, (b) the canonical UQFF expression involving $\rho_{\text{vac,SCm}}$ or $\rho_{\text{vac,UA}}$, (c) the observational target (Planck 2018, DES, eBOSS, JWST, etc.), and (d) the resulting predicted-vs-observed error.!

1 Anchor and Target

Observational anchor. ¡e.g. Planck 2018 $\Omega_{\Lambda}h^2 = \dots$ at $z = 0$!.

Reference. ¡DOI / arXiv / catalog ID! — this MUST be a real citation.

UQFF target quantity. ¡name and symbol!.

2 Canonical Equation

$$\boxed{< UQFFEQUATIONinvolving\rho_{\text{vac,SCm}}or\rho_{\text{vac,UA}} >} \quad (1)$$

The structural derivation enters via $\rho_{\text{vac,SCm}} = 4\sqrt{\pi} \cdot 10^{-37} \text{ J/m}^3$ under the G9 structural closure (PAPER_409, hydrogen geometry); the $|SO(5)| = 10$ ratio to $\rho_{\text{vac,UA}}$ is G7 (PAPER_1160).

3 Numerical Closure

Quantity	Predicted (UQFF v5.78)	Observed
¡quantity 1!	¡value!	¡value $\pm$ error!
¡quantity 2!	¡value!	¡value $\pm$ error!

Error. $\varepsilon = |P - O|/O = < Z > \%$.

4 Cross-references

- Canonical vacuum source: `dpm.vacuum.manifold.py` L97–98 (v3.0).
- Calculator: ¡e.g. `CondensedPhysics3.py::CosmologicalConstantCalculator`!
- Prior closure(s): PAPER_¡NN!, PAPER_¡MM!.

5 Reproducibility

```
from dpm_vacuum_manifold import RHO_VAC_SCM, RHO_VAC_UA
# show the one-liner that reproduces the predicted value
```

References

- [1] Planck Collaboration, *Planck 2018 results VI. Cosmological parameters*, A&A 641, A6 (2020).
arXiv:1807.06209.
- [2] PAPER_409 (Star-Magic): *Hydrogen-Geometry Derivation of $\rho_{\text{vac,SCM}}$* .
- [3] PAPER_1160 (Star-Magic): *F_TRZ G7 Structural Closure / —SO(5)— Ratio*.